

1 More convergence tests

Question 1. Determine whether the following series converge or diverge. State which test you used.

1. $\sum_{n=1}^{\infty} (-1)^n \cdot 2^{1/n}$

2. $\sum_{n=1}^{\infty} \frac{5 \cdot 2^{n+1}}{3^{n-1}}$

3. $\sum_{n=1}^{\infty} \cos(n) 2^{-(n^2)}$

4. $\sum_{n=1}^{\infty} \frac{\cos(\pi n)}{\sqrt{n}}$

5. $\sum_{n=1}^{\infty} \frac{n!}{\sqrt{(2n)!}}$

6. $\sum_{n=1}^{\infty} \left(\frac{5}{4}\right)^n \left(\frac{4n}{5n+1}\right)$

7. $\sum_{n=1}^{\infty} \sin^2\left(\frac{1}{n}\right)$

8. $\sum_{n=1}^{\infty} \sin^n\left(1 + \frac{1}{n}\right)$

9. $\sum_{n=1}^{\infty} \left(1 - \frac{1}{n}\right)^{(n^2)}$

10. $\sum_{n=1}^{\infty} \frac{n!}{n^n}$

Question 2. The following boxed argument **falsely concludes** that the series $\sum_{n=1}^{\infty} \frac{1+2(-1)^n}{n}$ converges. Identify the mistake(s) and provide a correct argument for the divergence of the series.

Notice that

$$\sum_{n=1}^{\infty} \frac{1+2(-1)^n}{n} = -\frac{1}{1} + \frac{3}{2} - \frac{1}{3} + \frac{3}{4} - \frac{1}{5} + \frac{3}{6} - \dots$$

is an alternating series and the limit of $-\frac{1}{n} \leq \frac{1+2(-1)^n}{n} \leq \frac{3}{n}$ as $n \rightarrow \infty$ is 0 by the sandwich theorem. Thus, it converges by the alternating series test.

Question 3 (More practice). Determine whether the following series converge or diverge. State which test you used.

1. $\sum_{n=1}^{\infty} \frac{1}{(\ln n)^3}$

2. $\sum_{n=1}^{\infty} \frac{2^{(n^2)}}{n!}$

3. $\sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{\sqrt{n}}\right)$

4. $\sum_{n=1}^{\infty} \frac{(\ln n)^n}{n}$

5. $\sum_{n=1}^{\infty} a_n$ where $a_n = \begin{cases} \frac{1}{n} & \text{if } n \text{ is a square number} \\ -\frac{1}{n^2} & \text{otherwise} \end{cases}$

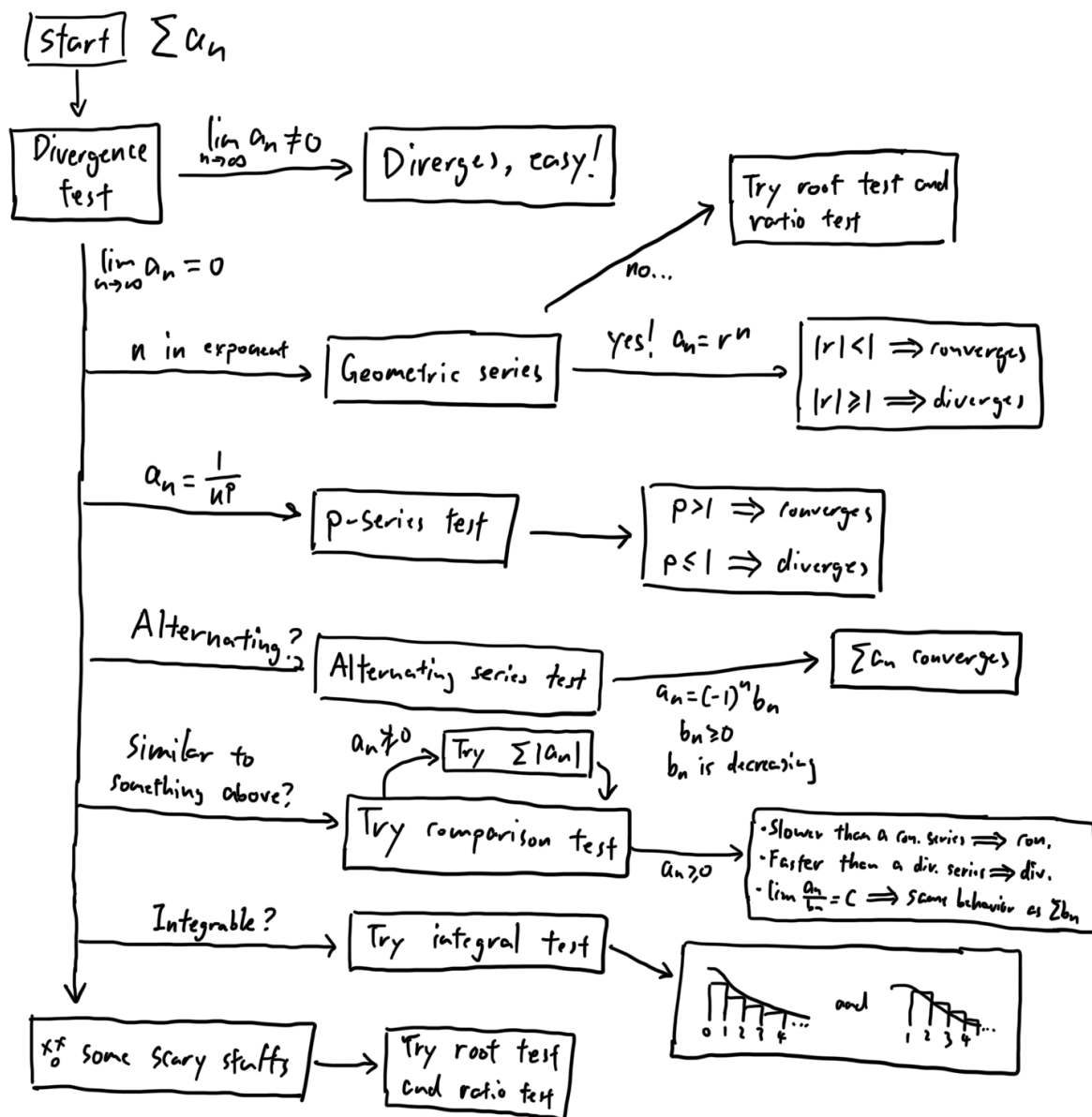


Diagram credit to Raymond Chou who TA-ed MAT 21C in Fall 2020.
 Don't forget telescoping series! But it is more of a technique for evaluating series than a convergence test.